CSE251

Assignment 01

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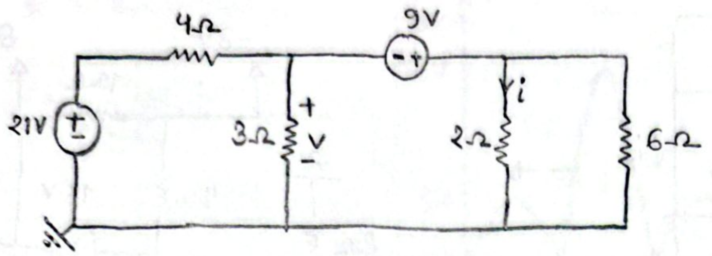
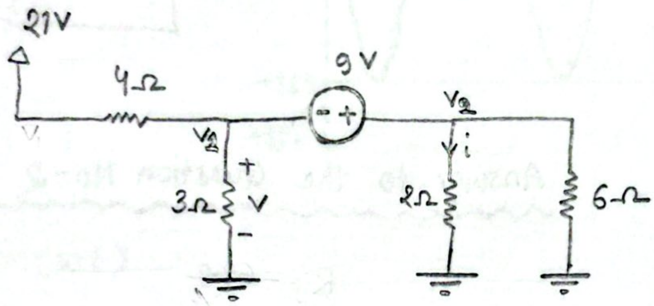
Answer to the Question No- 1



(a)



(b)

KCL on V_1 & V_2

$$\frac{V_1 - 21}{4} + \frac{V_1 - 0}{3} + \frac{V_2 - 0}{2} + \frac{V_2 - 0}{6} = 0$$

$$\Rightarrow 3V_1 - 63 + 4V_1 + 6V_2 + 2V_2 = 0$$

$$\Rightarrow 7V_1 + 8V_2 = 63 \quad \text{—(i)}$$

Supernode relation,

$$-V_1 + V_2 = 9 \quad \text{—(ii)}$$

Solving (i) & (ii)

$V_1 = -0.6\ V$

$V_2 = 8.4V$

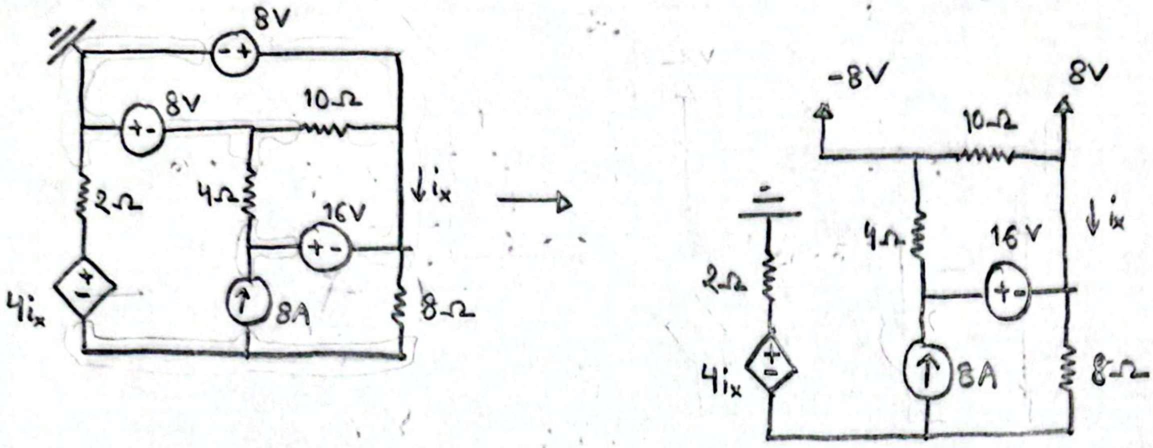
Now,

$$V = V_1 - 0$$
$$= -0.6 - 0$$
$$\boxed{\therefore V = -0.6\ V}$$

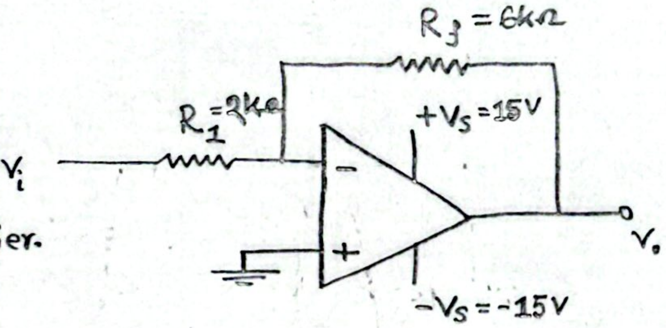
$$i = \frac{V_2 - 0}{2} = \frac{8.4 - 0}{2}$$

$$\boxed{\therefore i = 4.2\ A}$$

(c)



Answer to the Question No-2



Inverting Amplifier.

(a) $V_i = 4V$

We know, $V_o = -\frac{R_f}{R_1} V_i$

$= -\frac{6}{2} \times 4$

$= -12V$

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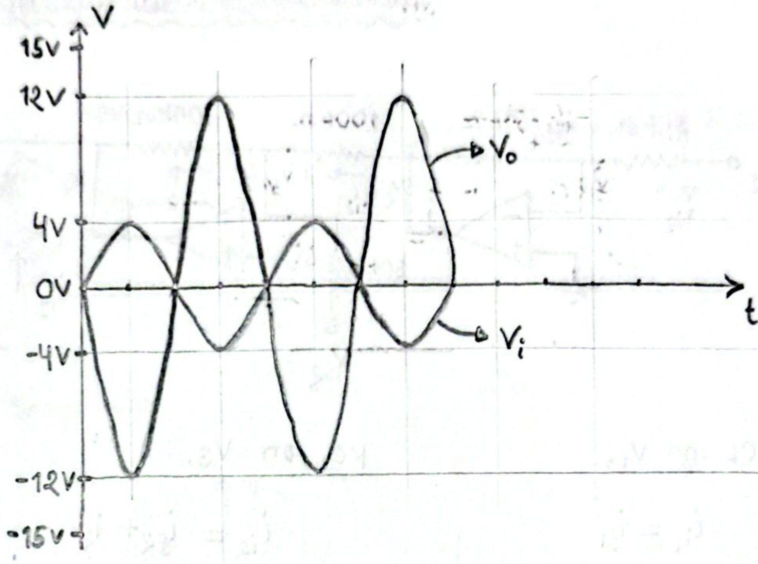
(b) $V_i = 4\sin(\omega t)$

$$V_o = -\frac{R_f}{R_1} V_i$$

$$= -\frac{6}{2} \times 4\sin(\omega t)$$

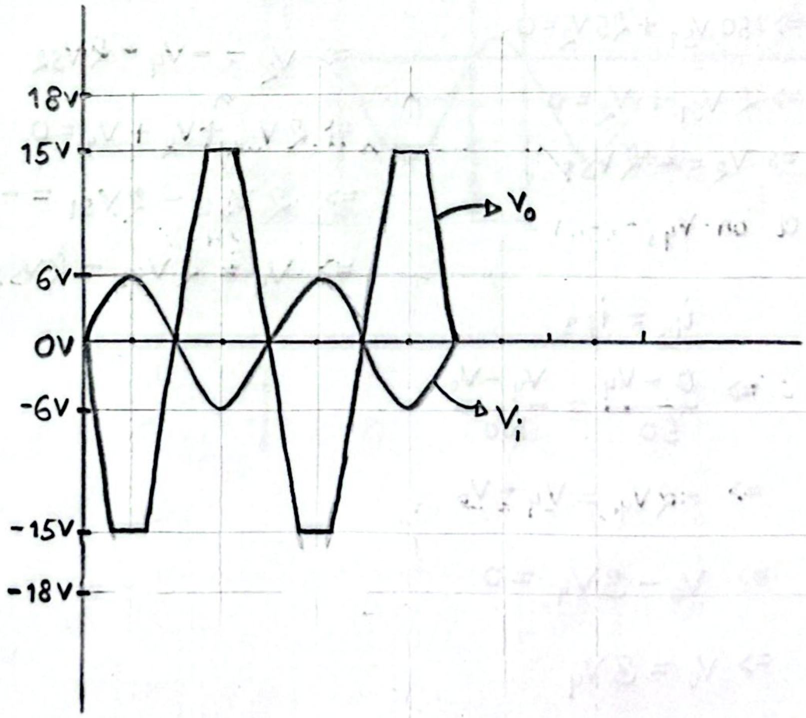
$$= -12\sin(\omega t)$$



(c) $V_i = 6\sin(\omega t)$

$$V_o = -\frac{6}{2} \times 6\sin(\omega t)$$

$$= -18\sin(\omega t)$$



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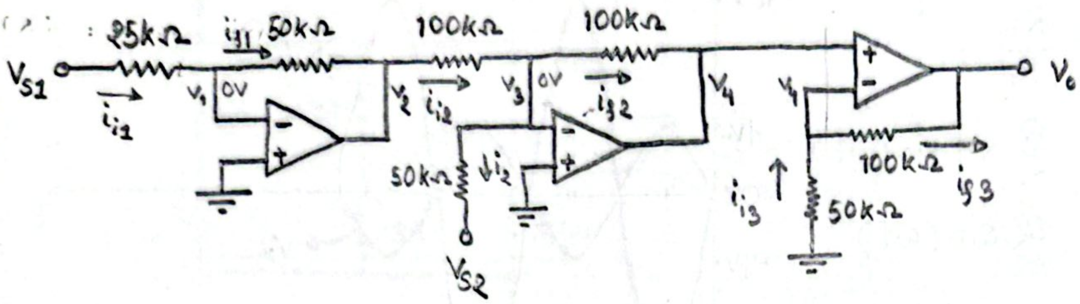
Answer to the Question No - 3



KCL on V_1,

$$i_{i1} = i_{f1}$$

$$\Rightarrow \frac{V_{S1}-0}{25} = \frac{0-V_2}{50}$$

$$\Rightarrow 50V_{S1} + 25V_2 = 0$$

$$\Rightarrow 2V_{S1} + V_2 = 0$$

$$\Rightarrow V_2 = -2V_{S1}$$

KCL on V_3,

$$i_{i2} = i_{f2} + i_2$$

$$\Rightarrow \frac{V_2-0}{100} = \frac{0-V_4}{100} + \frac{0-V_{S2}}{50}$$

$$\Rightarrow V_2 = -V_4 - 2V_{S2}$$

$$\Rightarrow 2V_{S2} + V_2 + V_4 = 0$$

$$\Rightarrow 2V_{S2} - 2V_{S1} = -V_4$$

$$\Rightarrow V_4 = 2V_{S1} - 2V_{S2}$$

KCL on V_4,

$$i_{i3} = i_{f3}$$

$$\Rightarrow \frac{0-V_4}{50} = \frac{V_4-V_o}{100}$$

$$\Rightarrow -2V_4 = V_4 - V_o$$

$$\Rightarrow V_o - 3V_4 = 0$$

$$\Rightarrow V_o = 3V_4$$

$$\Rightarrow V_o = 3(2V_{S1} - 2V_{S2})$$

$$\Rightarrow V_o = 6V_{S1} - 6V_{S2}$$

$$\therefore V_o = 6(V_{S1} - V_{S2})$$

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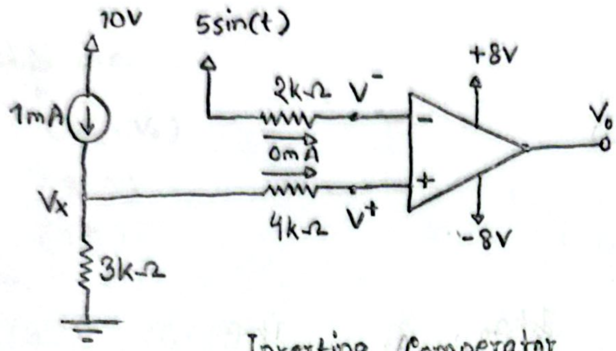
Answer to the Question No-4



10V (P-P) sine wave = 5 sin(t)

Inverting Comperator

KCL on V_x,

$$1 = \frac{V_x - 0}{3}$$

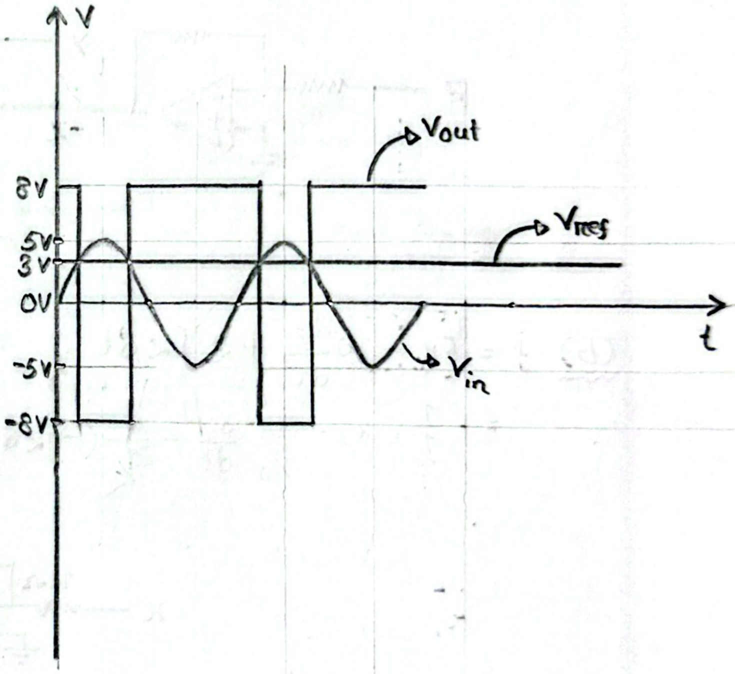
$$\Rightarrow \boxed{V_x = 3\,V}$$

Here,

$$V_{ref} = V^{+} = V_x = 3V$$

$$V_{in} = V^{-} = 5 \sin(t)$$

$$V_o = \begin{cases} 8V \ ; & V_{in} < V_{ref} \\ -8V \ ; & V_{in} > V_{ref} \end{cases}$$



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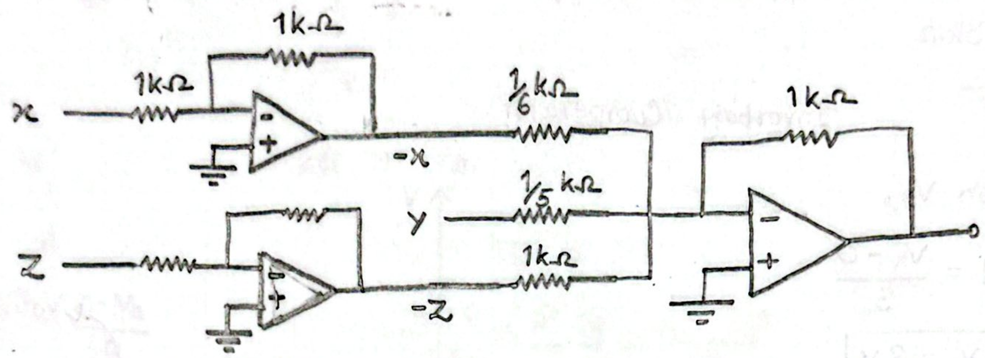
Answer to the Question No - 5

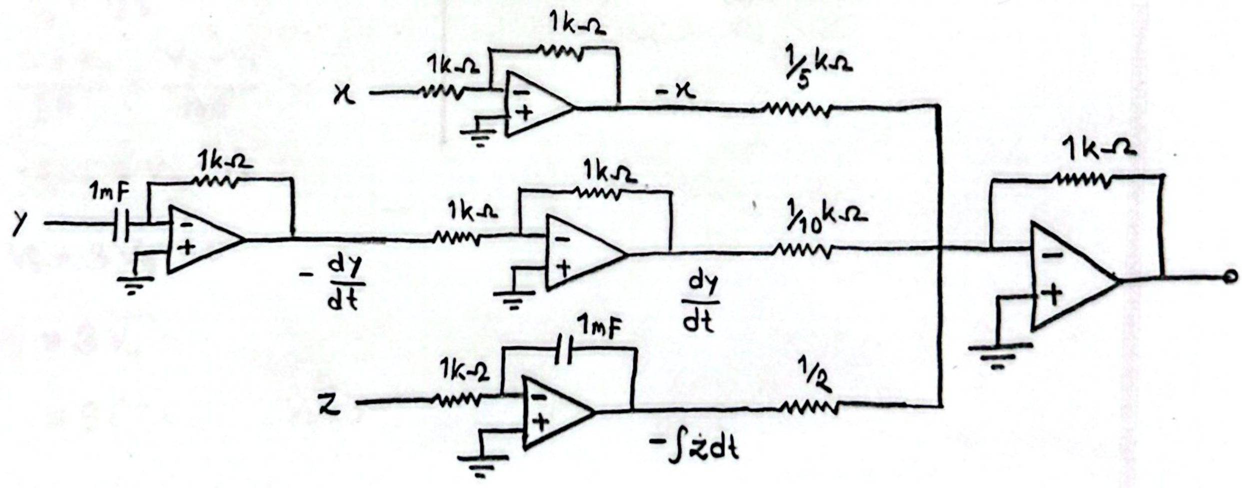
(a) $f = 6x - 5y + z$

$$= -\frac{1}{1/6}(-x) - \frac{1}{1/5}y - \frac{1}{1}(-z)$$



(b) $f = 5x - 10\frac{dy}{dt} + 2\int z\,dt$

$$= -\frac{1}{1/5}(-x) - \frac{1}{1/10}\frac{dy}{dt} - \frac{1}{1/2}\left(-\int z\,dt\right)$$



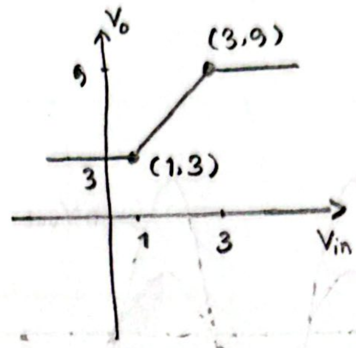
Answer to the Question No-6

Points are,

(V_{in}, V_o)

$(3, 9)$

$(1, 3)$



Non-Inverting Amplifier.

Now,

$$\frac{V_{in}-3}{3-1} = \frac{V_o-9}{9-3}$$

$\Rightarrow 2V_o - 18 = 6V_{in} - 18$

$\Rightarrow 2V_o = 6V_{in}$

$\Rightarrow V_o = \frac{6}{2} V_{in}$

$\Rightarrow V_o = 3V_{in}$

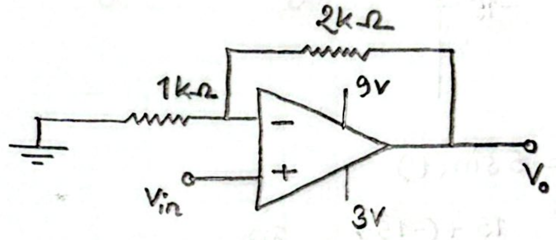
$\therefore V_o = (1+2) V_{in}$

Here, gain = (3)

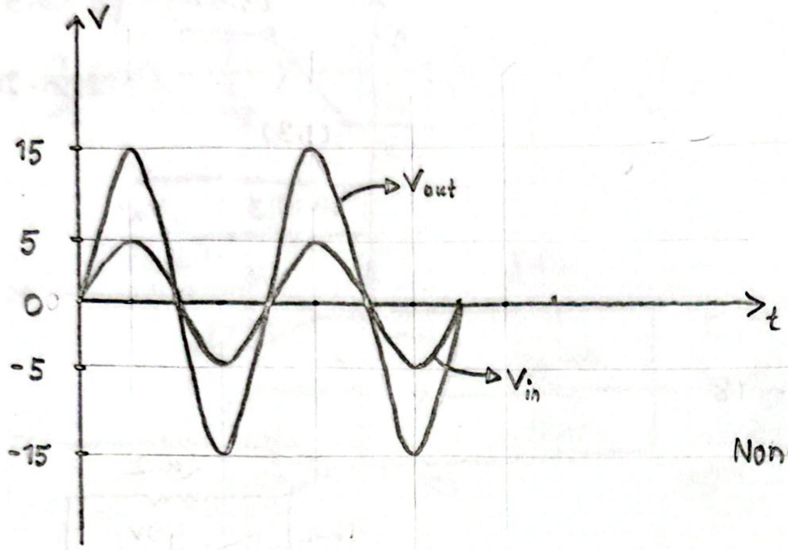
$\Rightarrow 1 + \frac{R_f}{R_1} = 1 + 2$

$\Rightarrow \frac{R_f}{R_1} = 2$

Let, $R_f = 2k\Omega$ & $R_1 = 1k\Omega$



Answer to the Question No-7



Non-Inverting /

$V_i = 5\sin(t)$

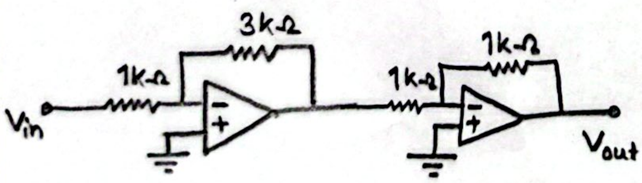
$$\text{gain} = \frac{15+(-15)}{5-(-5)} = \frac{30}{10} = 3$$

(i) Using inverting Amplifier,

$$\text{gain} = \frac{R_f}{R_1}$$

$$\Rightarrow 3 = \frac{R_f}{R_1}$$

let, $R_f = 3k\Omega$, & $R_1 = 1k\Omega$



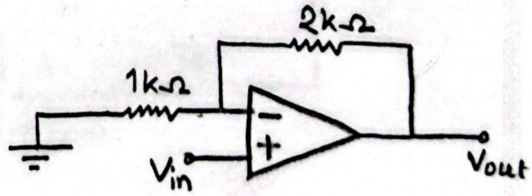
(ii) Using Non-Inverting Amplifier

$$\text{gain} = 1 + \frac{R_f}{R_1}$$

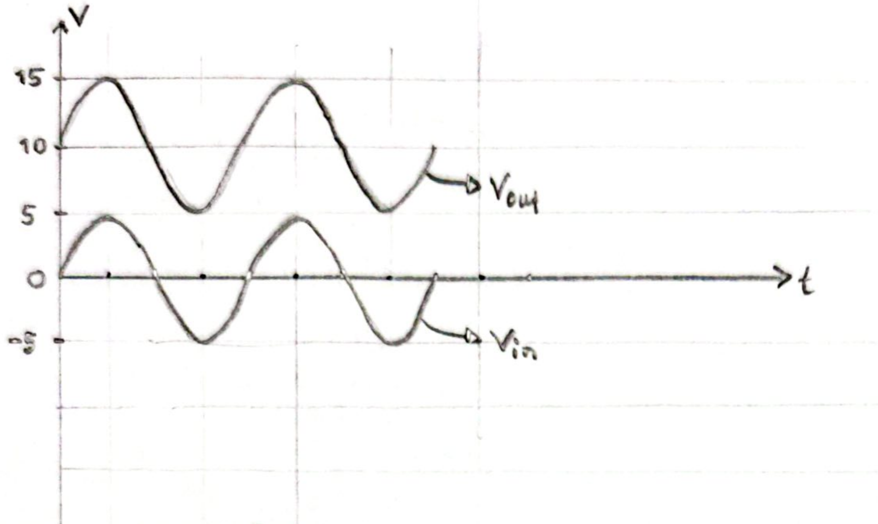
$$\Rightarrow 3 = 1 + \frac{R_f}{R_1}$$

$$\Rightarrow \frac{R_f}{R_1} = 2$$

let, $R_f = 2k\Omega$ & $R_1 = 1k\Omega$



Answer to the Question No-8



$V_{in} = 5\sin(t)$

$V_{out} = 10 + 5\sin(t)$